

The Nonstationary Twisted Spaces in Example 5.7 Are Not Nested

A negative answer to the printed twisted-wavelet question

Autonomous counterexample candidate for human review

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Status. Candidate full negative answer to the wavelet-existence question as printed, likely valid. The spaces in the source construction fail the first nesting axiom of a nonstationary multiresolution analysis. The separate conjecture on Riesz sequences of all twisted B-splines is not resolved.

1 Source construction and question

Das, Massopust, and Radha [1] define a nonstationary twisted multiresolution analysis by requiring, in particular,

$$V_j \subseteq V_{j+1} \quad (j \in \mathbb{Z}).$$

In Example 5.7 (source PDF pages 26–28), they take

$$\phi = \phi_1 = \mathbf{1}_{[0,1) \times [0,1)}, \quad N_j = 2^j \phi(2^j \cdot, 2^j \cdot), \quad \Phi_j = N_{2j},$$

and define V_j as the closed span of

$$g_{j,k,l} := D_{2^{-j}} \left(T_{(2^{-j}k, 2^{-j}l)}^t \right)^{2^{2j}} \Phi_j, \quad k, l \in \mathbb{Z}.$$

Here $D_a f(x, y) = af(ax, ay)$ and

$$(T_{(u,v)}^t)^\lambda f(x, y) = e^{\pi i \lambda (vx - uy)} f(x - u, y - v).$$

The source computes

$$g_{j,k,l}(x, y) = 2^j e^{\pi i (lx - ky)} \mathbf{1}_{[k, k+2^{-j}) \times [l, l+2^{-j})}(x, y), \quad (1)$$

then infers nesting from the fact that the level- $(j+1)$ support square is contained in the level- j square. This implication goes in the wrong direction.

At the bottom of page 28 the authors ask whether one can find Ψ_j whose twisted translates form a Riesz basis for W_j , where $V_j \oplus W_j = V_{j+1}$, and thereby obtain a Riesz basis for $L^2(\mathbb{R}^2)$.

2 Proof intuition

At every nonnegative level the generators occupy small squares anchored at integer lattice points. Increasing the level makes each occupied square smaller but does not refine the integer anchor lattice. Thus level $j+1$ leaves open gaps inside the supports occupied at level j . Closure of a span cannot create mass outside the union of the supports of its generators. A single level- j generator therefore witnesses the failure of nesting.

$$(T_{(k,l)}^\lambda)^{-1} f(x, y) := e^{-i\lambda(x-k)(y-l)} f(x-k, y-l), \quad (x, y) \in \mathbb{R}^2.$$

For the above definition and some properties of λ -twisted translations, we refer to [17].

Definition 5.6. A sequence of closed subspaces $\{V_j : j \in \mathbb{Z}\}$ of $L^2(\mathbb{R}^2)$ is said to generate a nonstationary twisted *multiresolution analysis* of $L^2(\mathbb{R}^2)$ if

- (i) $V_j \subset V_{j+1}, \forall j \in \mathbb{Z}$.
- (ii) $\overline{\bigcup_{j \in \mathbb{Z}} V_j} = L^2(\mathbb{R}^2)$ and $\bigcap_{j \in \mathbb{Z}} V_j = \{0\}$.
- (iii) For $j \in \mathbb{Z}$, there exists $\Phi_j \in V_j$ such that $\{D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Phi_j : k, l \in \mathbb{Z}\}$ is a *Riesz* basis for V_j .

We refer to the functions Φ_j are λ -twisted scaling functions at level j .

Then the usual arguments of *multiresolution analysis* lead to the fact that if there exists a function Ψ_j , for each $j \in \mathbb{Z}$, such that

$$(5.2) \quad \{D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Psi_j : k, l \in \mathbb{Z}\}$$

is a *Riesz* basis for W_j , then $\{D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Psi_j : j, k, l \in \mathbb{Z}\}$ is a *Riesz* basis for $L^2(\mathbb{R}^2)$. The elements of this collection at level j are termed λ -twisted wavelets.

Example 5.7. Let $\phi = \phi_1$. Let $\Phi_j = N_j$, where $N_j := 2^j \phi(2^j(\cdot, \cdot)), \forall j \in \mathbb{Z}$. Define $V_j := \text{span}\{D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Phi_j : k, l \in \mathbb{Z}\}, j \in \mathbb{Z}$. For $j \in \mathbb{Z}$, we have

$$D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Phi_j(x, y) = 2^j e^{\pi i(lx - ky)} \chi_{[k, k+2^{-j}] \times [l, l+2^{-j}]}(x, y).$$

Hence $[k, k+2^{-(j+1)}] \times [l, l+2^{-(j+1)}] \subset [k, k+2^{-j}] \times [l, l+2^{-j}]$. This leads to $V_j \subset V_{j+1}$. Let $f \in \bigcap_{j \in \mathbb{Z}} V_j$. Then $f(x, y) = c e^{\pi i(lx - ky)} \chi_{(-\infty, \infty) \times (-\infty, \infty)}(x, y)$, for some constant c . Then $|f(x, y)| = |c| \chi_{(-\infty, \infty) \times (-\infty, \infty)}(x, y)$. If $f \in L^2(\mathbb{R}^2)$, then $|f| \in L^2(\mathbb{R}^2)$, which forces $c = 0$ and which in turn implies $f = 0$, proving that $\bigcap_{j \in \mathbb{Z}} V_j = \{0\}$. An element in $\bigcup_{j \in \mathbb{Z}} V_j$ can be written as a product of

functions $g \cdot u$, where $|g| = 1$ and u belongs to the class of piecewise constant functions on $\mathbb{R}^2$. Hence $\bigcup_{j \in \mathbb{Z}} V_j$ contains the class of piecewise constant functions on $\mathbb{R}^2$ which implies that $\overline{\bigcup_{j \in \mathbb{Z}} V_j} = L^2(\mathbb{R}^2)$.

We claim that $\{D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Phi_j : k, l \in \mathbb{Z}\}$ is a *Riesz* basis for V_j for each fixed $j \in \mathbb{Z}$. To this end, let $\{\alpha_{k,l}\} \in \ell^2(\mathbb{Z}^2)$. We need to prove that there exist $A, B > 0$ such that

$$A \|\{\alpha_{k,l}\}\|_{\ell^2(\mathbb{Z}^2)}^2 \leq \left\| \sum_{k,l \in \mathbb{Z}} \alpha_{k,l} D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Phi_j \right\|^2 \leq B \|\{\alpha_{k,l}\}\|_{\ell^2(\mathbb{Z}^2)}^2.$$

Figure 1: Definition 5.6 and the construction and support formula in Example 5.7, source PDF page 26.

3 Negative answer

For $j \geq 0$, set

$$E_j = \bigcup_{k,l \in \mathbb{Z}} [k, k+2^{-j}] \times [l, l+2^{-j}].$$

Theorem 1. For the spaces in Example 5.7,

$$V_j \not\subset V_{j+1} \quad \text{for every } j \geq 0.$$

In particular there is no subspace W_0 with $V_0 \oplus W_0 = V_1$. Hence no family $(\Psi_j)_{j \in \mathbb{Z}}$ satisfying the open question exists for the spaces as defined.

Proof. Equation (1) shows that every level- j generator is supported in E_j . Therefore every finite linear combination is supported in E_j . The subspace

$$L^2(E_j) = \{f \in L^2(\mathbb{R}^2) : f = 0 \text{ a.e. on } \mathbb{R}^2 \setminus E_j\}$$

$$\begin{aligned}
|R| &\leq 2\|\{\alpha_{k,l}\}\|_{\ell^2(\mathbb{Z}^2)}^2 \sum_{(r,s) \in B_j^{(1)}} |\langle T_{(r,s)}^t N_j, N_j \rangle| \\
(5.6) \quad &\leq \left(\frac{2^{2j+2}}{\pi} \#(B_j^{(1)}) \right) \|\{\alpha_{k,l}\}\|_{\ell^2(\mathbb{Z}^2)}^2.
\end{aligned}$$

But $B_j^{(1)}$ is the set of pairs (r, s) , where both r and s are odd. Thus $\#(B_j^{(1)}) = 2^{-2j-1}$. Hence, (5.6) gives $|R| \leq \frac{2}{\pi} \|\{\alpha_{k,l}\}\|_{\ell^2(\mathbb{Z}^2)}^2$, from which it follows that

$$\left(1 - \frac{2}{\pi}\right) \|\{\alpha_{k,l}\}\|_{\ell^2(\mathbb{Z}^2)}^2 \leq S \leq \left(1 + \frac{2}{\pi}\right) \|\{\alpha_{k,l}\}\|_{\ell^2(\mathbb{Z}^2)}^2,$$

proving our assertion. Thus, if one can find a function Ψ_j , for each $j \in \mathbb{Z}$ in $L^2(\mathbb{R}^2)$, such that

$$\{D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Psi_j : k, l \in \mathbb{Z}\}$$

is a *Riesz* basis for W_j ($V_j \oplus W_j = V_{j+1}$), then

$$\{D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Psi_j : j, k, l \in \mathbb{Z}\}$$

is a *Riesz* basis for $L^2(\mathbb{R}^2)$. We leave it as an open question to the interested reader.

In Figure 2 below, we have depicted $D_{2^{-j}}(T_{(2^{-j}k, 2^{-j}l)}^t)^{2^{2j}} \Phi_j \in V_j$ for two triples (j, k, l) .

Figure 2: The open wavelet-existence question, source PDF page 28.

is closed, so

$$V_j \subseteq L^2(E_j). \quad (2)$$

Fix $j \geq 0$ and take the level- j generator with $k = l = 0$:

$$g_{j,0,0} = 2^j \mathbf{1}_{[0, 2^{-j}) \times [0, 2^{-j})} \in V_j.$$

Consider the positive-measure rectangle

$$R_j = (2^{-(j+1)}, 2^{-j}) \times (0, 2^{-(j+1)}).$$

Because $j \geq 0$, the only level- $(j+1)$ square meeting the open unit cell $(0, 1) \times (0, 1)$ is $[0, 2^{-(j+1)}) \times [0, 2^{-(j+1)})$. Consequently $R_j \cap E_{j+1} = \emptyset$. But $g_{j,0,0} = 2^j$ throughout R_j . It follows from (2) at level $j+1$ that $g_{j,0,0} \notin V_{j+1}$. Hence $V_j \not\subseteq V_{j+1}$.

If $V_0 \oplus W_0 = V_1$ for any subspace W_0 , then necessarily $V_0 \subseteq V_1$, contradicting the case $j = 0$. Thus the requested W_0 and Ψ_0 do not exist, and neither can the requested all-level family. $\square$

4 Direct algebra check and scope

For completeness, the displayed support formula follows directly from the source definitions. Substitution gives

$$\begin{aligned}
g_{j,k,l}(x, y) &= 2^{-j} e^{\pi i 2^{2j} ((2^{-j}l)(2^{-j}x) - (2^{-j}k)(2^{-j}y))} \\
&\quad \times 2^{2j} \phi(2^{2j}(2^{-j}x - 2^{-j}k), 2^{2j}(2^{-j}y - 2^{-j}l)) \\
&= 2^j e^{\pi i (lx - ky)} \mathbf{1}_{[k, k+2^{-j}) \times [l, l+2^{-j})}(x, y).
\end{aligned}$$

Thus the counterexample does not depend on an OCR reading or a numerical plot.

This resolves the literal existence question for Example 5.7, not every possible corrected construction. One could repair the support defect by changing the scale/translation convention so that the anchor lattice refines with the support, but that would define different spaces and a new problem. The all-orders Riesz-sequence conjecture in Section 4 is also separate.

5 Verification and novelty status

The proof uses only supports and closedness of $L^2(E_j)$. The decisive review point is whether the published formula (1) and the definition of V_j are intended literally. A bounded search of the run indexes, arXiv, the 2022 journal record, and later papers citing the source found no explicit correction or resolution of this exact wavelet question. Priority is therefore not asserted; the mathematical status is a candidate counterexample pending human review.

References

- [1] S. R. Das, P. Massopust, and R. Radha, *Twisted B-splines in the complex plane*, Appl. Comput. Harmon. Anal. 56 (2022), 250–282; arXiv:2009.04146.